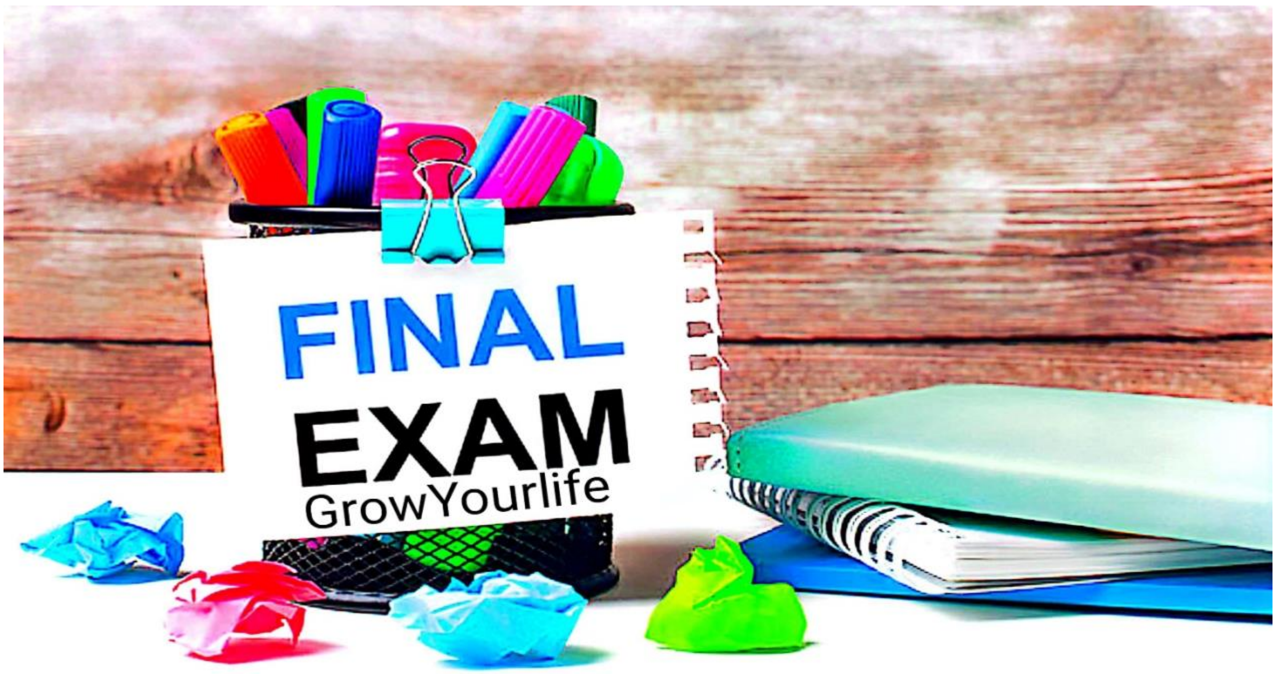
Cs201P

Subjective File

Error Based Questions

Special Final term Mega File

GrowYourLife VULMS



1. Missing Header File

Statement: Identify and correct the issue in the following code.

cpp
```
........
int main() {
    cout << "Hello, CS201!";
    return 0;
}
```

Error: The header file <iostream> is missing.
Corrected Code:

cpp
....
```
#include <iostream>
using namespace std;
int main() {
    cout << "Hello, CS201!";
    return 0;
}
```

---

2. Missing Semicolon

Statement: Find and fix the syntax error in this code.

cpp
....
```
int main() {
    int num = 10
    cout << "Number: " << num;
    return 0;
}
```

Error: Missing semicolon after int num = 10.
Corrected Code:

cpp
....
```
int main() {
    int num = 10;
    cout << "Number: " << num;
    return 0;
}
```

---

3. Incorrect Syntax in if Condition

Statement: Debug the following code.

cpp
....
```
int main() {
```

```
    int x = 10;
    if x > 5 {
        cout << "x is greater than 5";
    }
    return 0;
}
```

Error: Parentheses are missing around the condition in the if statement.
Corrected Code:

cpp
....

```
int main() {
    int x = 10;
    if (x > 5) {
        cout << "x is greater than 5";
    }
    return 0;
}
```

4.Variable Not Declared

Statement: Correct the undeclared variable error in the code.

cpp
....

```
int main() {
    cout << num;
    return 0;
}
```

Error: The variable num is not declared.
Corrected Code:

cpp
....

```
int main() {
    int num = 20;
    cout << num;
    return 0;
}
```

---

5. Missing Return Statement

Statement: Correct the following code.

cpp
....
```
int main() {
    cout << "Program is running.";
}
```

Error: The main function must return an integer.
Corrected Code:

cpp
....
```
int main() {
    cout << "Program is running.";
    return 0;
}
```

---

6. Logical Error in if-else

Statement: Debug the logic error in this code.

cpp
....
```
int main() {
    int a = 5, b = 10;
    if (a > b)
        cout << "a is greater";
    else
        cout << "b is greater";
    return 0;
}
```

Error: The condition if (a > b) will evaluate as false, so check the variable values.
Corrected Code: *(Logic remains correct; no change needed)*

7.Misuse of Assignment Operator in if

Statement: Identify and fix the issue.

cpp
....
```
int main() {
    int a = 5;
    if (a = 10) {
        cout << "a is 10";
    }
    return 0;
}
```

Error: The assignment operator = is used instead of the equality operator ==.
Corrected Code:

cpp
....
```
int main() {
    int a = 5;
    if (a == 10) {
        cout << "a is 10";
    }
    return 0;
}
```

---

8. Infinite Loop

Statement: Fix the infinite loop.

cpp
....
```
int main() {
    int i = 0;
    while (i < 5) {
        cout << i << " ";
    }
    return 0;
```

}

Error: The value of i is not incremented inside the loop, causing an infinite loop.
Corrected Code:

cpp
....
```
int main() {
    int i = 0;
    while (i < 5) {
        cout << i << " ";
        i++;
    }
    return 0;
}
```

---

9. Array Index Out of Bound

Statement: Correct the array access issue.

cpp
....
```
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    cout << arr[5];
    return 0;
}
```

Error: The array index 5 is out of bounds.
Corrected Code:

cpp
....
```
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    cout << arr[4];
    return 0;
}
```

10.Wrong Data Type for cin

Statement: Fix the input error in the following code.

cpp
....
```
int main() {
    int age;
    cout << "Enter your age: ";
    cin >> "age";
    return 0;
}
```

Error: cin is used with a string literal "age" instead of the variable age.
Corrected Code:

cpp
....
```
int main() {
    int age;
    cout << "Enter your age: ";
    cin >> age;
    return 0;
}
```

11. Array Size Mismatch

Statement: Identify the issue in the array initialization.

cpp
....
```
int main() {
    int arr[3] = {1, 2, 3, 4};
    return 0;
}
```

Error: The array size is declared as 3, but 4 elements are provided.
Corrected Code:

cpp
....

```
int main() {
    int arr[4] = {1, 2, 3, 4};
    return 0;
}
```

---

12. Using Undeclared Functions

Statement: Debug the following code.

cpp
....

```
int main() {
    int result = add(5, 3);
    cout << "Result: " << result;
    return 0;
}
```

Error: The function add is not declared or defined.
Corrected Code:

cpp
....

```
int add(int a, int b) {
    return a + b;
}

int main() {
    int result = add(5, 3);
    cout << "Result: " << result;
    return 0;
}
```

---

13. Type Mismatch in Function Arguments

Statement: Fix the type mismatch in the code.

cpp
....

```
void printAge(int age) {
```

```
    cout << "Age: " << age;
}

int main() {
    printAge("25");
    return 0;
}
```

Error: The argument "25" is a string, but the function expects an integer.
Corrected Code:

cpp
....
```
int main() {
    printAge(25);
    return 0;
}
```

14.Accessing Private Members

Statement: Find the issue with accessing private members.

cpp
....
```
class Student {
    int age;
};

int main() {
    Student s;
    s.age = 20;
    return 0;
}
```

Error: age is private and cannot be accessed directly.
Corrected Code:

cpp
....
```
class Student {
private:
```

```cpp
    int age;
public:
    void setAge(int a) { age = a; }
};

int main() {
    Student s;
    s.setAge(20);
    return 0;
}
```

---

15. Missing Header for sqrt

Statement: Debug the following code.

cpp
....
```cpp
int main() {
    double result = sqrt(25);
    cout << "Result: " << result;
    return 0;
}
```

Error: #include <cmath> is missing for the sqrt function.
Corrected Code:

cpp
....
```cpp
#include <cmath>
int main() {
    double result = sqrt(25);
    cout << "Result: " << result;
    return 0;
}
```

---

16. Divide by Zero

Statement: Correct the runtime error in the code.

cpp
....
```
int main() {
    int a = 10, b = 0;
    cout << "Result: " << a / b;
    return 0;
}
```

Error: Division by zero causes a runtime error.
Corrected Code:

cpp
....
```
int main() {
    int a = 10, b = 0;
    if (b != 0)
        cout << "Result: " << a / b;
    else
        cout << "Division by zero error!";
    return 0;
}
```

17.Unreachable Code

Statement: Debug the code with unreachable statements.

cpp
....
```
int main() {
    return 0;
    cout << "This line will not execute.";
}
```

Error: Code after return 0; is unreachable.
Corrected Code:

cpp
....
```
int main() {
    cout << "This line will execute.";
    return 0;
```

}

---

18. Conflicting Return Types

Statement: Fix the return type issue.

cpp
....
```
void add(int a, int b) {
    return a + b;
}
```

Error: The function has a void return type but attempts to return a value.
Corrected Code:

cpp
....
```
int add(int a, int b) {
    return a + b;
}
```

---

19. Null Pointer Dereference

Statement: Find and fix the null pointer error.

cpp
....
```
int main() {
    int *ptr = nullptr;
    cout << *ptr;
    return 0;
}
```

Error: Dereferencing a null pointer causes undefined behavior.
Corrected Code:

cpp
....
```
int main() {
```

```
    int value = 10;
    int *ptr = &value;
    cout << *ptr;
    return 0;
}
```

20. Incorrect Use of new and delete

Statement: Correct memory allocation issues.

cpp
....
```
int main() {
    int *arr = new int[5];
    // Missing delete[]
    return 0;
}
```

Error: Memory allocated with new[] must be deallocated with delete[].
Corrected Code:

cpp
....
```
int main() {
    int *arr = new int[5];
    delete[] arr;
    return 0;
}
```

---

21. File Not Closed

Statement: Fix the file handling issue.

cpp
....
```
#include <fstream>
int main() {
    ofstream file("output.txt");
    file << "Hello, World!";
    // Missing file.close();
```

```
    return 0;
}
```

Error: The file is not explicitly closed.
Corrected Code:

cpp
....
```
#include <fstream>
int main() {
    ofstream file("output.txt");
    file << "Hello, World!";
    file.close();
    return 0;
}
```

---

22. Logical Error in Loop

Statement: Fix the infinite loop.

cpp
....
```
int main() {
    int i = 0;
    while (i < 5) {
        cout << i;
        // Missing increment of i
    }
    return 0;
}
```

Error: The loop never terminates due to the missing increment.
Corrected Code:

cpp
....
```
int main() {
    int i = 0;
    while (i < 5) {
        cout << i;
```

```
        i++;
    }
    return 0;
}
```

23. Recursive Function Without Base Case

Statement: Debug the following code.

cpp
....
```
int factorial(int n) {
    return n * factorial(n - 1);
}
```

Error: Missing base case causes infinite recursion.
Corrected Code:

cpp
....
```
int factorial(int n) {
    if (n <= 1) return 1;
    return n * factorial(n - 1);
}
```

---

24. Missing break in switch

Statement: Fix the fall-through issue in the code.

cpp
....
```
int main() {
    int choice = 2;
    switch (choice) {
        case 1:
            cout << "One";
        case 2:
            cout << "Two";
    }
    return 0;
```

}

Error: Missing break statements in switch cases.
Corrected Code:

cpp
....
```
int main() {
    int choice = 2;
    switch (choice) {
        case 1:
            cout << "One";
            break;
        case 2:
            cout << "Two";
            break;
    }
    return 0;
}
```

---

25. Shadowing Variables

Statement: Debug the shadowing issue.

cpp
....
```
int x = 10;
int main() {
    int x = 20;
    cout << x;
    return 0;
}
```

Error: Local variable x shadows the global variable x.
Corrected Code: *(No correction needed unless global x is required.)*

26. Missing default in a switch Statement

Statement: Fix the incomplete switch statement.

cpp
....
```
int main() {
    int option = 3;
    switch (option) {
        case 1:
            cout << "Option 1 selected";
            break;
        case 2:
            cout << "Option 2 selected";
            break;
    }
    return 0;
}
```

Error: The switch statement does not handle unexpected values of option.
Corrected Code:

cpp
....
```
int main() {
    int option = 3;
    switch (option) {
        case 1:
            cout << "Option 1 selected";
            break;
        case 2:
            cout << "Option 2 selected";
            break;
        default:
            cout << "Invalid option selected";
    }
    return 0;
}
```

---

27. Incorrect Use of const Variables Without Initialization

Statement: Debug the use of a const variable.

cpp
....
```
int main() {
    const int x;
    x = 10;
    cout << x;
    return 0;
}
```

Error: A const variable must be initialized at the time of declaration.
Corrected Code:

cpp
....
```
int main() {
    const int x = 10;
    cout << x;
    return 0;
}
```

---

28. Implicit Conversion Issue

Statement: Correct the implicit conversion problem.

cpp
....
```
int main() {
    float num = 5.75;
    int result = num;
    cout << result;
    return 0;
}
```

Error: The implicit conversion from float to int causes data loss.
Corrected Code:

cpp
....
```
int main() {
    float num = 5.75;
```

```
    int result = static_cast<int>(num);
    cout << result;  // Output: 5
    return 0;
}
```

29. Memory Leak by Not Deleting Allocated Memory

Statement: Fix the memory leak in the following code.

cpp
....

```
int main() {
    int *arr = new int[10];
    arr[0] = 5;
    cout << arr[0];
    return 0;  // Memory allocated to `arr` is not freed
}
```

Error: The memory allocated using new[] is not released, leading to a memory leak.
Corrected Code:

cpp
....

```
int main() {
    int *arr = new int[10];
    arr[0] = 5;
    cout << arr[0];
    delete[] arr;  // Properly free the allocated memory
    return 0;
}
```

---

30. Misusing Pointer Arithmetic

Statement: Debug the pointer arithmetic issue in the following code.

cpp
....

```
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
```

```
    int *ptr = arr;
    cout << *(ptr + 5);  // Accessing out of bounds
    return 0;
}
```

Error: Accessing *(ptr + 5) is out of bounds and causes undefined behavior.
Corrected Code:

```
cpp
........
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    int *ptr = arr;
    cout << *(ptr + 4);  // Correctly access the last element of the array
    return 0;
}
```